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Ex.No. 14: 5G Throughput vs SNR with Modulation Labels

OBJECTIVE 1:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
% SNR values (dB)
```

```
SNR = 0:5:30;
```

```
% Maximum Throughput (Mbps)
```

```
Max_TP = 100;
```

```
% Throughput calculation
```

```
Throughput = Max_TP * (1 - exp(-SNR/10));
```

```
% Display values
```

```
disp('SNR(dB) Throughput(Mbps)')
```

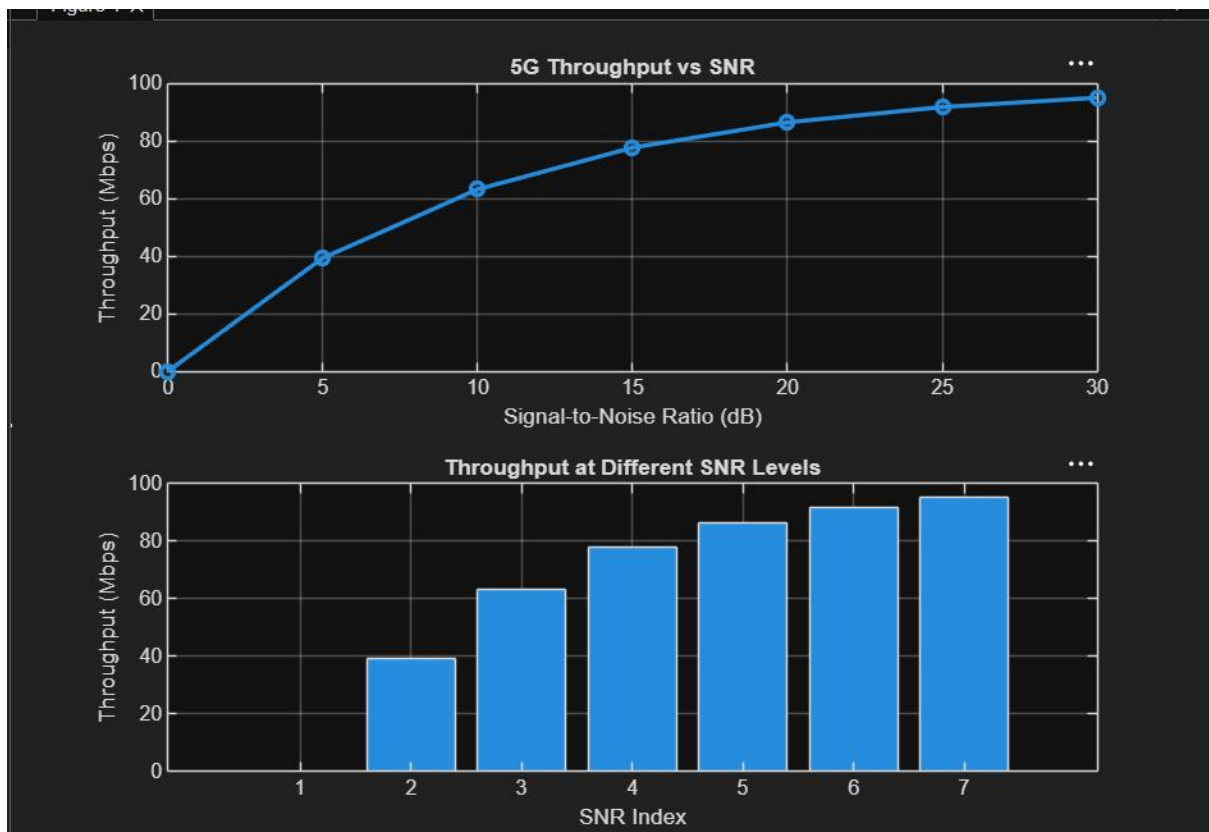
```
disp([SNR' Throughput'])
```

```
figure
```

```
% ----- Graph 1 -----  
subplot(2,1,1)  
plot(SNR, Throughput, '-o', 'LineWidth', 2)  
title('5G Throughput vs SNR')  
xlabel('Signal-to-Noise Ratio (dB)')  
ylabel('Throughput (Mbps)')  
grid on
```

```
% ----- Graph 2 -----  
subplot(2,1,2)  
bar(Throughput)  
title('Throughput at Different SNR Levels')  
xlabel('SNR Index')  
ylabel('Throughput (Mbps)')  
grid on
```

0



OBJECTIVE 2:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
% SNR values (dB)
```

```
SNR = 0:5:30;
```

```
% Simulated Bit Error Rate (BER)
```

```
BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];
```

```
% Reliability (%)
```

```
Reliability = (1 - BER) * 100;
```

```
% Display values
```

```
disp('SNR(dB) BER Reliability(%)' )
```

```
disp([SNR' BER' Reliability'])
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
semilogy(SNR, BER, '-o', 'LineWidth', 2)
```

```
title('SNR vs Bit Error Rate in 5G')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Bit Error Rate (BER)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(SNR, Reliability, '-s', 'LineWidth', 2)
```

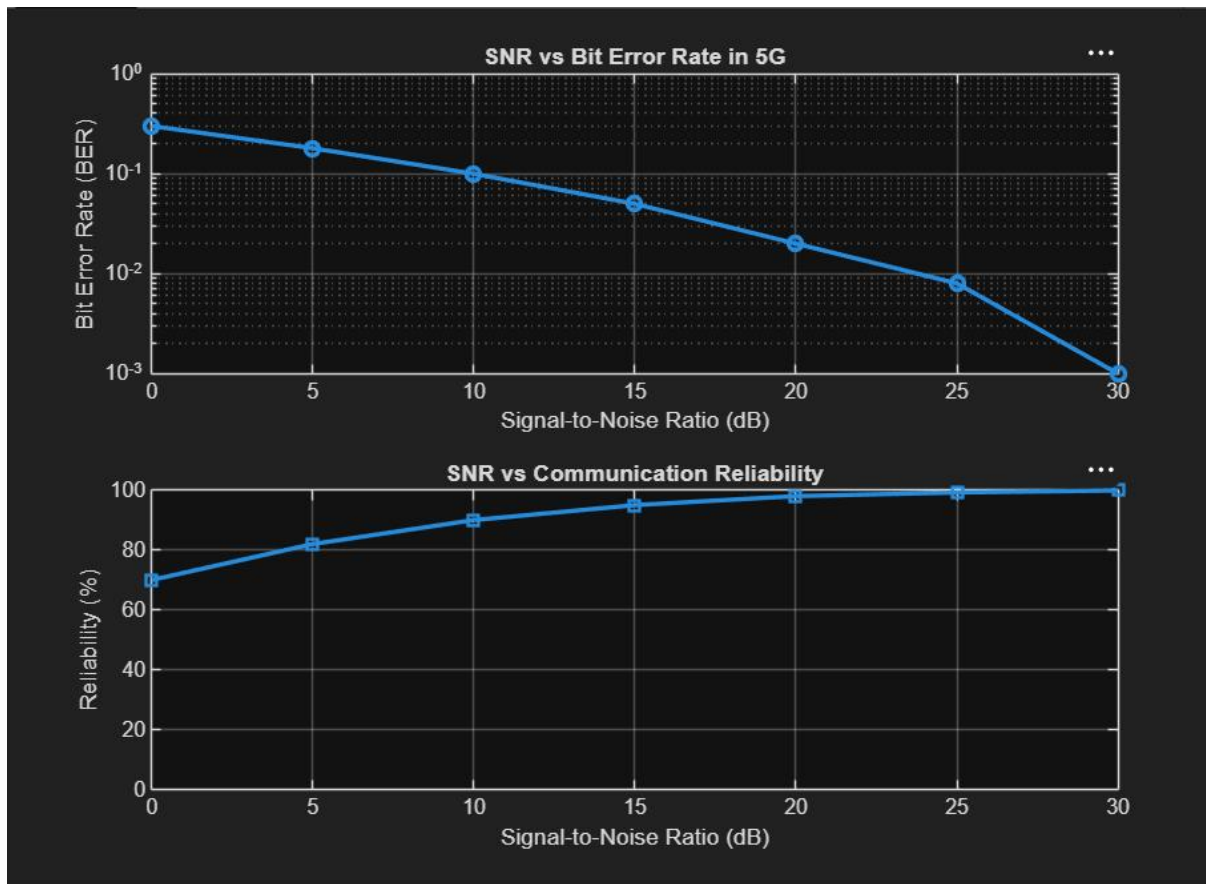
```
title('SNR vs Communication Reliability')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Reliability (%)')
```

```
grid on
```

ylim([0 100])



OBJECTIVE3:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Modulation schemes
```

```
Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};
```

```
% Required SNR (dB)
```

```
SNR = [5 10 18 25];
```

```
% Spectral Efficiency (bits/s/Hz)
```

```
Efficiency = [2 4 6 8];
```

```
% Display values
```

```
disp('Required SNR (dB) and Spectral Efficiency')
```

```
disp(table(Modulation', SNR', Efficiency', ...
```

```
    'VariableNames', {'Modulation','SNR_dB','Efficiency'}))
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(1:4, SNR, '-s', 'LineWidth', 2)
```

```
xticks(1:4)
```

```
xticklabels(Modulation)
```

```
title('Required SNR for Different Modulation Schemes')
```

```
xlabel('Modulation Scheme')
```

```
ylabel('Required SNR (dB)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(SNR, Efficiency, '-o', 'LineWidth', 2)
title('SNR vs Spectral Efficiency')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Spectral Efficiency (bits/s/Hz)')
grid on
```

